DSAA3073: Theories in Data Science

Tutorial 5A: Reuse Data without Reusing the Final Answer

150-minute core tutorial - 15-minute break - 60-minute protected buffer

Wei WANG · HKUST(GZ) · 2026-08-28

Explorative projection

LESSON BACKBONE

Ridge gives a family of estimators indexed by λ. With limited data, how can we compare that family honestly, and when can leave-one-out evaluation be recovered from one fit?

Tutorial 4B ended before choosing λ. It gave us the ridge estimator and the warning that training fit cannot answer the prediction question. We now assign distinct jobs to fitting, selection, and final evaluation. That separation leads from a holdout set to repeated folds and finally to leave-one-out cross-validation.

The last step seems to demand one fit per observation. For fixed-design ordinary and ridge least squares, however, the full-data residual and one diagonal entry of a matrix contain exactly the required deletion effect. Deriving that identity—and knowing when not to use it—is the mathematical destination of the lesson.

The three core blocks take **50, 35, and 65 minutes**. Take the scheduled 15-minute break after the first block. The final 60 minutes are protected for questions, reconstruction, and recovery; they introduce no required knowledge.

Clock	Mathematical work	Protected time
0-50	Separate data roles; construct holdout and k-fold estimates	50 min core
50-65	Scheduled break	15 min
65-100	Specialize the fold construction to leave-one-out and identify its target	35 min core
100-165	Derive the ridge hat matrix and exact PRESS residual identity; audit scope	65 min core
165-225	Questions, recovery, alternative calculations	60 min buffer

Three jobs for one finite dataset

Tutorial 4B made exactly three items available here, in this order:

Available item	What we may do with it
empirical risk minimization	Identify the training-fit baseline and distinguish it from population performance.
ridge regression	Use $\hat{\beta}_\lambda = (X^T X + \lambda I)^{-1} X^T y$ for a declared $\lambda > 0$.
structural risk minimization	Recognize fit-versus-complexity control without treating every selection rule as the same bound.

IN-CLASS CHECK

Prompt: A ridge fit has the smallest training error at $\lambda = 0$. Which of the three available items explains that calculation, and why does none of them yet certify the best predictive λ?

Hint: Name the scored data before interpreting a minimum.

Answer:

> ERM explains why minimizing an unpenalized training criterion favors the best fit on those observations. Ridge supplies the candidate family, and SRM supplies a fit-complexity perspective. None provides fresh out-of-sample evidence for the numerical value of λ in this data analysis, so the predictive comparison remains open.

Suppose the available observations are divided before any score is examined. The training subset fits each candidate. A validation subset compares the already-declared candidates. A final test subset is sealed until every model, hyperparameter, threshold, and stopping decision is fixed.

The two pipelines below use the same observations but make different claims. Trace every backward arrow before reading the diagnosis.

path	fit candidates		choose λ		report performance
valid	training data	$\rightarrow$	validation data	$\rightarrow$	sealed final test
contaminated	training data	$\rightarrow$	final test	$\leftarrow$	minimum final-test error

Green path: information moves forward once. Red path: the reported data moved backward into the choice it is then asked to evaluate.

Table 1: Data-role audit. Circle every dataset whose values can change the selected λ; only the first path leaves final evidence untouched.

Definition: Holdout validation

Holdout validation fits every candidate on a training subset and measures its declared loss on a disjoint validation subset. For a predictor fixed independently of that validation subset, the validation mean estimates its fresh-data risk. Choosing the smallest among many validation means is a separate selection operation, so that selected minimum is not an untouched final-performance estimate.

Let T, V, and E denote the training, validation, and final-evaluation index sets. For squared loss and a candidate λ fixed before inspecting V,

$$\hat{R}_{V(\lambda)} = \frac{1}{|V|}\sum_{i\in V}\left(y_i - x_i^T\hat{\beta}_{\lambda,T}\right)^2, \tag{1}$$

where $\hat{\beta}_{\lambda,T}$ uses only observations in T. Conditional on the trained predictor, independent validation observations make this an ordinary sample mean of fresh losses. If the same values choose $\hat{\lambda} = \operatorname{argmin}_\lambda \hat{R}_{V(\lambda)}$, then $\hat{R}_{V(\hat{\lambda})}$ is a selected minimum. It remains useful for choosing; it is not promoted to final evidence.

Worked example: Audit a ridge-selection report

A team predeclares $\lambda \in \{0.1, 1, 10\}$. Training fits give validation MSEs $0.44, 0.31, 0.36$, so the team chooses $\lambda = 1$, refits on training plus validation data, and evaluates once on the sealed test set. This is a valid selection workflow.

Reporting 0.31 as the final test MSE would be wrong: it is the smallest of three reused validation outcomes and evaluates fits trained on fewer data than the final refit. The final test result evaluates the completed procedure.

IN-CLASS CHECK

Prompt: A team tries 40 values of λ and reports the smallest validation MSE as its final error. Give two distinct reasons the report overclaims.

Hint: Separate selection reuse from the training-size change after refitting.

Answer:

First, the validation outcomes selected the winner, so their minimum is optimistic as an estimate of the complete selection procedure's performance. Second, the final estimator is normally refit on more data than any candidate used in that validation comparison. An untouched test set, new data, or an outer evaluation layer is needed for final evidence.

One split preserves independence but can spend too much data on a single validation role. Repeated folds let every observation take that role once.

Definition: k-fold cross-validation

Partition the available **selection sample** into disjoint folds $F_1, \ldots, F_k$. For candidate λ, fit the learning algorithm k times, each time excluding one fold, and average the held-out losses:

$$\mathrm{CV}_{k(\lambda)} = \frac{1}{n} \sum_{j=1}^{k} \sum_{i \in F_j} \ell\Big(y_i, f_{-F_j}^{(\lambda)}(x_i)\Big). \tag{2}$$

Every scored observation must be excluded from the fit that produces its prediction. After choosing λ, refit with that value on the complete selection sample; do not bring the final test set into the fold table.

For a five-fold ridge comparison, the invariant is easier to see in a ledger than in a verbal recipe.

row	scored fold	fit uses	candidate	loss producer
1	F_1	$F_2 \cup \ldots \cup F_5$	λ	$f_{-F_1}^{(\lambda)}$
2	F_2	$F_1 \cup F_3 \cup \ldots \cup F_5$	λ	$f_{-F_2}^{(\lambda)}$
3	F_3	**fill this cell**	λ	**fill this cell**
4	F_4	$F_1 \cup F_2 \cup F_3 \cup F_5$	λ	$f_{-F_4}^{(\lambda)}$
5	F_5	$F_1 \cup \ldots \cup F_4$	λ	$f_{-F_5}^{(\lambda)}$

Table 2: Fold ledger. Complete row 3, then check that each observation is scored exactly once and never by a fit that saw its fold.

Row 3 uses $F_1 \cup F_2 \cup F_4 \cup F_5$, and its loss producer is $f_{-F_3}^{(\lambda)}$. The same fold partition is reused across candidates so that candidate comparisons do not add avoidable partition noise.

IN-CLASS CHECK

Prompt: For folds of sizes $3, 2, 3$, write $\mathrm{CV}_3(\lambda)$ using the eight individual held-out losses. Why is an unweighted average of the three fold means wrong here?

Hint: The definition averages observations, not fold labels.

Answer:

> If $L_i^{(\lambda)}$ is observation i's loss from the fit excluding its fold, then $\mathrm{CV}_3(\lambda) = \sum_{i=1}^{8} \frac{L_i^{(\lambda)}}{8}$. An unweighted average of the three fold means gives the two-observation fold the same weight as either three-observation fold, so it does not equal the mean held-out loss over observations.

Larger k changes several things at once: each fit uses more observations, there are more fits, and the training sets overlap more. The fold losses are therefore dependent. Without additional assumptions about the learner, distribution, and stability, there is no universal monotone bias-variance ranking in k, and averaging k folds does not automatically reduce a standard deviation by $\sqrt{k}$.

IN-CLASS CHECK

Prompt: A classmate says, "Ten folds divide variance by ten because they are ten averages." Identify the missing premise and name two other quantities that change with k.

Hint: Inspect the training sets behind the fold losses.

Answer:

> Dividing variance by ten would require an independence or covariance calculation that overlapping fold fits do not supply. Increasing k also increases the training size in each fit and the number of fits; it changes overlap, computation, and potentially the algorithmic target. No universal ranking follows from the fold count alone.

Let every observation take one turn outside

Set $k = n$ and let fold F_i contain only observation i. The fold ledger now has n rows, and every fit uses exactly $n - 1$ observations.

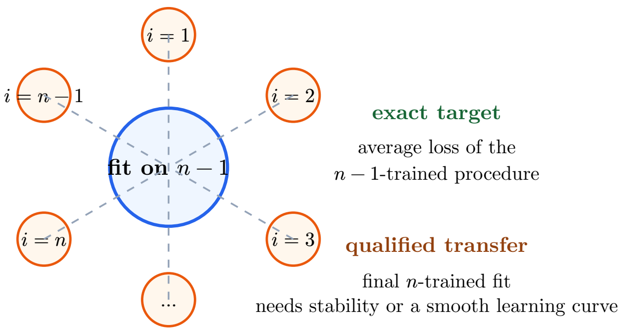

Figure 1: Exclusion wheel. Trace one orange observation into scoring only after the central fit has excluded it; then distinguish the exact $n - 1$ target from the qualified n-sample transfer.

Definition: Leave-one-out cross-validation

Leave-one-out cross-validation (LOOCV) is k-fold cross-validation with $k = n$. If $f_{-i}^{(\lambda)}$ is trained on all observations except i, then for squared loss

$$\mathrm{LOOCV}(\lambda) = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - f_{-i}^{(\lambda)}(x_i)\right)^2. \tag{3}$$

Each summand evaluates an algorithm trained on $n - 1$ observations. The same candidate value and feature construction are used in every deletion fit.

For the fixed observed dataset, the singleton folds are determined, so there is no randomness from choosing a fold partition. That does **not** remove sampling variability: another dataset drawn from the population produces a different LOOCV value. Nor does the definition make the $n - 1$ and n training targets identical. Their proximity requires stability or a sufficiently smooth learning curve near sample size n.

Worked example: What does the score evaluate?

With $n = 6$, every deleted ridge fit uses five observations. The six squared errors therefore measure how the declared five-observation training procedure predicts an omitted case, averaged over the six omissions. After selecting λ, the final ridge fit uses all six selection observations. Calling the LOOCV score exactly unbiased for that six-observation fit would skip the training-size change.

IN-CLASS CHECK

Prompt: For a fixed dataset of size 80, state (i) the training size behind each LOOCV prediction, (ii) what is deterministic, and (iii) what remains random under repeated sampling.

Hint: Do not use "deterministic" as a synonym for "zero variance."

Answer:

Each prediction comes from a fit trained on 79 observations. The list of singleton folds and the computed score are deterministic conditional on this fixed dataset and algorithm. Across new i.i.d. datasets, both the fits and the score vary. Applying the result to the final 80-observation fit additionally needs a stability or smooth-learning-curve qualification.

For one candidate value, naive LOOCV calls the fitting algorithm n times. That count creates the next question: does ridge carry enough algebraic structure to recover every deleted prediction from its full-data fit?

Send the ridge estimator back to the response space

Recall the registered coefficient estimator from Chapter 4:

$$\hat{\beta}_\lambda = (X^T X + \lambda I_d)^{-1} X^T y, \quad X \in \mathbb{R}^{n\times d}, \quad \lambda > 0. \tag{4}$$

Multiplying by X produces fitted responses rather than coefficients:

$$\hat{y} = X\hat{\beta}_\lambda = \underbrace{X(X^T X + \lambda I_d)^{-1} X^T}_{H_\lambda} y. \tag{5}$$

response $y \in \mathbb{R}^n$	$\to$	coefficients $\hat{\beta}_\lambda \in \mathbb{R}^d$	$\to$	fitted response $\hat{y} = H_\lambda y \in \mathbb{R}^n$

First arrow: $(X^T X + \lambda I_d)^{-1} X^T$. Second arrow: X. Their composition is H_λ.

Table 3: Response-to-fit matrix map. Check all three dimensions before naming the diagonal influence used by PRESS.

Definition: Linear smoother and ridge hat matrix

A fitted-response rule is a **linear smoother** when $\hat{y} = Hy$ for a matrix H that is fixed with respect to the response vector y. For fixed design and a fixed positive ridge parameter, the registered ridge hat matrix is

$$H_\lambda = X(X^T X + \lambda I_d)^{-1} X^T. \tag{6}$$

Its diagonal entry $h_{\lambda,ii} = (H_\lambda)_{ii}$ is the ridge leverage of observation i: the coefficient multiplying y_i in its own full-data fitted value. Plain H_λ is a matrix; $\mathcal{H}$ remains the course notation for a hypothesis class.

The dimensions are worth checking once. The product is $n \times d$, then $d \times d$, then $d \times n$, so $H_\lambda \in \mathbb{R}^{n \times n}$ and $H_\lambda y \in \mathbb{R}^n$. The matrix depends on X and the fixed λ, not on y.

IN-CLASS CHECK

Prompt: Let $X \in \mathbb{R}^{40\times 6}$. Give the dimensions of $(X^T X + \lambda I)^{-1}$ and H_λ, then explain why changing y does not require rebuilding H_λ.

Hint: Follow the matrix map from response to coefficients and back.

Answer:

> The inverse is 6×6 and H_λ is 40×40. With X and λ fixed, every factor defining H_λ is unchanged when the response values change; only the final multiplication $H_\lambda y$ changes.

For ordinary least squares with full column rank, $H_0 = X(X^T X)^{-1} X^T$ is an orthogonal projector and $\mathrm{tr}(H_0) = \mathrm{rank}(X)$. For positive ridge,

$$\mathrm{tr}(H_\lambda) = \sum_j \frac{\mu_j}{\mu_j + \lambda} \tag{7}$$

over positive Gram eigenvalues μ_j. This trace is the ridge **effective degrees of freedom** and is generally smaller than the unregularized rank. The OLS rank identity and the ridge effective-dimension statement are not one universal trace-equals-d rule.

Before seeing any deletion formula, work only with quantities returned by one full ridge fit. For $\lambda = 1$, suppose

$$X = \mathrm{diag}\left(\frac{1}{2}, 1, 2, \frac{1}{3}\right), \quad A_1^{-1} = (X^T X + I_4)^{-1} = \mathrm{diag}\left(\frac{4}{5}, \frac{1}{2}, \frac{1}{5}, \frac{9}{10}\right), \tag{8}$$

and the complete residual vector is $e = (0.40, -0.30, 0.12, -0.18)^T$. Because $H_1 = XA_1^{-1}X^T$, its diagonal entries can be calculated as $h_{1,ii} = x_i^T A_1^{-1} x_i$ without knowing a leave-one-out identity.

i	**row** x_i^T	e_i	$h_{1,ii}$	$1 - h_{1,ii}$
1	$\left(\frac{1}{2}, 0, 0, 0\right)$	0.40	**blank**	**blank**
2	$(0, 1, 0, 0)$	-0.30	**blank**	**blank**
3	$(0, 0, 2, 0)$	0.12	**blank**	**blank**
4	$\left(0, 0, 0, \frac{1}{3}\right)$	-0.18	**blank**	**blank**

Table 4: Full-fit leverage ledger. Compute both blank columns from the displayed inputs; no deleted residual has yet been supplied.

Commit before the reveal: Write all four leverage/complement pairs, then predict which observation will receive the greatest multiplicative residual inflation after deletion. Keep that prediction visible while reading the derivation.

Working space - show the four diagonal products and one prediction with a reason.

Delete one row without losing the labels

Write x_i^T for row i of X, and set

$$A_\lambda = X^T X + \lambda I_d, \quad \hat{y}_i = x_i^T \hat{\beta}_\lambda, \quad e_i = y_i - \hat{y}_i. \tag{9}$$

Deleting observation i changes the ridge normal equations to

$$(A_\lambda - x_i x_i^T)\hat{\beta}_{-i,\lambda} = X^T y - x_i y_i. \tag{10}$$

The Sherman-Morrison rank-one inverse identity gives

$$(A_\lambda - x_i x_i^T)^{-1} = A_\lambda^{-1} + \frac{A_\lambda^{-1} x_i x_i^T A_\lambda^{-1}}{1 - x_i^T A_\lambda^{-1} x_i}, \tag{11}$$

provided the denominator is nonzero. Since $x_i^T A_\lambda^{-1} x_i = h_{\lambda,ii}$, substitution and collection of the full-data residual yield

$$\hat{\beta}_{-i,\lambda} = \hat{\beta}_\lambda - \frac{A_\lambda^{-1} x_i e_i}{1 - h_{\lambda,ii}}. \tag{12}$$

Multiplying by x_i^T gives the deleted prediction and residual:

$$\hat{y}_{-i,i} = \hat{y}_i - \frac{h_{\lambda,ii} e_i}{1 - h_{\lambda,ii}}, \tag{13}$$

$$e_{-i,i} := y_i - \hat{y}_{-i,i} = \frac{e_i}{1 - h_{\lambda,ii}}. \tag{14}$$

Definition: Exact PRESS leave-one-out shortcut

For fixed-design ordinary or ridge least squares with the same feature construction and the same penalty in every deletion fit, the leave-one-out residual is

$$e_{-i,i} = \frac{y_i - \hat{y}_i}{1 - h_{\lambda,ii}}, \tag{15}$$

whenever $1 - h_{\lambda,ii} \neq 0$. Therefore

$$\mathrm{LOOCV}(\lambda) = \frac{1}{n} \sum_{i=1}^{n} \left(\frac{e_i}{1 - h_{\lambda,ii}} \right)^2. \tag{16}$$

The prediction sum of squares (PRESS) is the corresponding unaveraged sum. One compatible full-data fit supplies every residual and diagonal leverage.

The first commitment gives $h_{1,ii} = (0.20, 0.50, 0.80, 0.10)$ and $1 - h_{1,ii} = (0.80, 0.50, 0.20, 0.90)$. Observation 3 has the smallest complement, so it receives the largest multiplicative inflation. Now use the revealed identity to complete a second ledger; do not read the answer below until every deleted residual and the final PRESS sum are on your page.

i	e_i	$h_{\lambda,ii}$	$1-h_{\lambda,ii}$	$e_{-i,i}$
1	0.40	0.20	0.80	?
2	−0.30	0.50	0.50	?
3	0.12	0.80	0.20	?
4	−0.18	0.10	0.90	?

Table 5: PRESS ledger. The same full-data residual can have a very different deleted residual when the observation has large leverage.

Second commitment: Calculate all four deleted residuals, then fill both total boxes before continuing.

PRESS =		**LOOCV =**	

The completed deleted residuals are $0.50, -0.60, 0.60, -0.20$. Hence $\text{PRESS} = 0.25 + 0.36 + 0.36 + 0.04 = 1.01$ and $\text{LOOCV} = \frac{1.01}{4} = 0.2525$. Observation 3 has the smallest full-data residual magnitude but ties for the largest deleted residual magnitude because its leverage is high.

IN-CLASS CHECK

Prompt: For full residuals $(0.2, -0.4)$ and ridge leverages $(0.25, 0.5)$, compute both deleted residuals and the two-observation LOOCV squared-error score.

Hint: Correct each residual before squaring and averaging.

Answer:

The deleted residuals are $\frac{0.2}{0.75} = \frac{4}{15}$ and $-\frac{0.4}{0.5} = -0.8$. The score is $\frac{(\frac{4}{15})^2 + 0.8^2}{2} = \frac{\frac{16}{225} + \frac{16}{25}}{2} = \frac{16}{45} \approx 0.356$.

The matrix map is not the deletion theorem

The proof used more than the display $\hat{y} = Hy$. It used the compatible rank-one change in ordinary or quadratic-penalty least-squares normal equations, kept λ and the features fixed, and required a nonzero denominator. Full-data response-linearity alone does not supply those facts for an arbitrary learning algorithm.

Ordinary k-nearest-neighbor prediction is the key nonexample. With a fixed neighbor graph its in-sample predictions can be written as a weighted average of entries of y. But deleting observation i can change neighbor sets and weights, and the ordinary prediction convention may or may not include the query point itself. There is no corresponding least-squares rank-one identity. Therefore we do not apply the PRESS residual formula to ordinary k-NN.

Kernel ridge regression has a compatible quadratic structure, but the objects need to be explicit. Given training inputs $x_1, \ldots, x_n$, let the kernel Gram matrix $K \in \mathbb{R}^{n \times n}$ have entries $K_{ij} = k(x_i, x_j)$. Under the same scaling convention used above, the **kernel-ridge smoother** maps responses to fitted responses by

$$M_\lambda = K(K + \lambda I_n)^{-1}, \quad \hat{y} = M_\lambda y. \tag{17}$$

For the linear kernel $k(x, z) = x^T z$, we have $K = XX^T$. The push-through identity then gives

$$M_\lambda = XX^{T(XX^T+\lambda I_n)^{-1}} = X(X^TX + \lambda I_d)^{-1}X^T = H_\lambda. \tag{18}$$

Mohri et al., Exercise 11.9, therefore supplies compatibility evidence for efficient kernel-ridge leave-one-out computation, with ordinary ridge as this linear-kernel case. It is not the source of the exact formula proved here: the current fixed-design ridge PRESS identity follows from the normal equations and rank-one derivation above. Compatibility—not the phrase "linear in y" by itself—is what licenses the comparison.

IN-CLASS CHECK

Prompt: Classify each statement as sufficient or insufficient for the displayed PRESS identity: (a) $\hat{y} = Hy$ for a fixed full dataset; (b) ridge with fixed X, fixed λ, and nonzero $1 - h_{\lambda,ii}$; (c) ordinary k-NN whose neighbor set may change after deletion.

Hint: Ask which claim reproduces the deleted fit, not merely the full fit.

Answer:

> (a) is insufficient: it describes the full-data response map but not its case-deletion behavior. (b) is sufficient because the quadratic normal equations give the rank-one deletion identity under the stated conditions. (c) is insufficient; changing the dataset can change neighbors and weights, so ordinary k-NN is not assigned this PRESS formula.

Standard language to carry forward

Term or notation	Meaning in this tutorial	Recall cue
holdout validation	Fit on one subset; compare fixed candidates on a disjoint subset.	selection data are not final test data
k-fold cross-validation	Each observation is scored once by a fit that excludes its fold.	$\mathrm{CV}_k = \frac{1}{n}\sum$ held-out losses
leave-one-out cross-validation	The $k = n$ case; every prediction comes from an $n-1$-observation fit.	$\mathrm{LOOCV} = \frac{1}{n}\sum_i e_{-i,i}^2$
ridge hat matrix	The registered response map $\hat{y} = H_\lambda y$.	$H_\lambda = X(X^T X + \lambda I)^{-1} X^T$
PRESS shortcut	For compatible fixed-design least squares, correct each full residual by one minus its leverage.	$e_{-i,i} = \frac{e_i}{1-h_{\lambda,ii}}$

Reconstruct what Tutorial 5B may assume

Tutorial 5B may rely on exactly the following nine items, in this order:

No.	Available item	Productive recall
1	empirical risk minimization	Separate a training-fit minimum from final performance.
2	ridge regression	Use the registered estimator for fixed $\lambda > 0$.
3	structural risk minimization	Recognize a justified fit-complexity bound, not a validation score.
4	holdout validation	Separate fixed-candidate estimation, selection reuse, and final evaluation.
5	k-fold cross-validation	Exclude every scored fold from the fit producing its loss.
6	leave-one-out cross-validation	State the $n-1$ target and qualify transfer to the n-sample fit.
7	linear smoother and hat matrix	Decode $\hat{y} = H_\lambda y$ and diagonal leverage.
8	exact LOOCV shortcut	Use $\frac{e_i}{1-h_{\lambda,ii}}$ only with the compatible deletion identity.
9	H_λ notation	Keep plain H_λ distinct from hypothesis class $\mathcal{H}$.

IN-CLASS CHECK

Prompt: Without looking back, reconstruct the full nine-item chain: name the three incoming ideas, the three validation designs, the response-space matrix, the exact residual correction, and the registered matrix notation. Then say what remains unresolved.

Hint: The last sentence should create the need for Tutorial 5B without using its formulas.

Answer:

> The chain is ERM, ridge, SRM; holdout, k-fold, and leave-one-out validation; the linear smoother or hat matrix; the exact compatible least-squares LOOCV shortcut; and the notation H_λ. These tools compare held-out predictive losses, but they do not yet tell us whether a single-fit likelihood correction answers the same question or how its assumptions and target should be compared with validation and SRM.

Optional self-study appendix

OPTIONAL DEEP DIVE - 0 CLASS MINUTES

Question: What is generalized cross-validation, and why is it not another exact PRESS identity?

Answer.

For a declared linear smoother $\hat{y} = H_\lambda y$, generalized cross-validation (GCV) replaces the observation-specific denominators by their average:

$$\mathrm{GCV}(\lambda) = \frac{\frac{\|y - H_\lambda y\|^2}{n}}{\left(1 - \operatorname{tr} \frac{H_\lambda}{n}\right)^2}. \tag{19}$$

It uses effective degrees of freedom through $\operatorname{tr}(H_\lambda)$ and is useful under the smoother assumptions for which that average correction is justified. Unlike PRESS, it generally does not equal the sum of the exact deleted residuals when the leverages differ. It is an optional approximation here, exports no knowledge to Tutorial 5B, and consumes zero protected class minutes.

Post-class or self-study. Not a prerequisite for the core path.

Source note

The historical Week 5 Learning Sheet fixes the official scope of holdout, k-fold, LOOCV, and efficient computation, but its universal $\sqrt{k}$-variance reduction, monotone bias-versus-k table, historical LOOCV standard-error formula, generic-linear-smoother shortcut, ordinary-k-NN claim, and trace-equals-dimension wording are not reused.

The train-validation-test roles and the k-fold algorithm follow Shalev-Shwartz and Ben-David, Sections 11.2.1-11.2.5, printed/PDF pp. 146-150. The validation-versus-SRM boundary and finite-candidate independence argument are checked against Mohri et al., Sections 4.4-4.5, printed pp. 68-72 (PDF pp. 85-89). The n-versus-$n-1$ LOOCV qualification follows Claeskens and Hjort, Section 2.9, printed pp. 51-55 (PDF pp. 69-73).

Claeskens and Hjort, printed p. 55 (PDF p. 73), is the direct source for the ordinary-least-squares PRESS residual identity and its linear-mean, constant-variance scope. The ridge formula is derived above from the registered ridge estimator with fixed design and fixed penalty. Mohri, Rostamizadeh, and Talwalkar, Exercise 11.9, printed pp. 293-294 (PDF pp. 310-311), independently establishes efficient leave-one-out computation for kernel ridge regression; the displayed equality above identifies ordinary ridge only as its linear-kernel special case. No claim is made that arbitrary response-linearity, ordinary k-NN, or an undeclared smoother identity is sufficient.

DSAA3073: Theories in Data Science

Tutorial 5B: Match the Selection Rule to the Inferential Goal

150-minute core tutorial - 15-minute break - 60-minute protected buffer

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Explorative projection

LESSON BACKBONE

Validation scores held-out loss, while information criteria correct likelihood optimism. When do these tools answer related questions, when do AIC, BIC, and SRM answer different questions, and what evidence remains honest after selection?

Tutorial 5A gave us data-splitting rules and an exact leave-one-out shortcut for compatible least-squares fits. Those tools still leave a practical opening: a regular parametric likelihood contains information about how much its maximized training fit has adapted to the sample. We will turn that observation into AIC, then derive BIC from a different target.

The distinction matters. AIC corrects toward predictive conditional log score. BIC is a large-sample marginal-likelihood approximation and supports model identification only under a declared regime. Neither formula is a distribution-free PAC bound, and neither turns data already used for selection into final performance evidence.

The three core blocks take **50, 45, and 55 minutes**. Take the scheduled 15-minute break after the first block. The final 60 minutes are protected for questions, reconstruction, and recovery; they introduce no required knowledge.

Clock	Mathematical work	Protected time
0-50	Derive AIC as a predictive conditional-log-score correction and audit its sign and parameter count	50 min core
50-65	Scheduled break	15 min
65-110	Derive BIC from marginal likelihood; separate prediction, identification, and PAC claims	45 min core
110-165	Place selection inside and evaluation outside; choose a method from a complete study brief	55 min core
165-225	Questions, recovery, alternative calculations	60 min buffer

Correct optimism toward a predictive target

Tutorial 5A made exactly nine items available, in this order. They are not a topic list: each one has a job in the argument below.

No.	Available item	Use in this Tutorial
1	empirical risk minimization	Recognize why an in-sample optimum is not automatically a population-performance certificate.
2	ridge regression	Keep a concrete fitted family indexed by a tuning parameter.
3	structural risk minimization	Recall one genuine fit-plus-bound construction without transferring its guarantee.
4	holdout validation	Distinguish a fixed-candidate score, selection evidence, and final evaluation.
5	k-fold cross-validation	Score held-out losses from fits that exclude the scored fold.
6	leave-one-out cross-validation	Relate fresh conditional log score to one-case deletion under stated asymptotics.
7	linear smoother and hat matrix	Separate a response-space map from a likelihood parameter count.
8	exact LOOCV shortcut	Use it only for its compatible deletion identity, not as an information criterion.
9	H_λ notation	Keep the ridge matrix distinct from the hypothesis class $\mathcal{H}$.

IN-CLASS CHECK

Prompt: For ridge regression, which available item computes an exact compatible LOOCV score, which one describes the training optimization, and which one supplies a high-probability fit-complexity comparison? Why are those three answers not interchangeable?

Hint: Name the scored object and guarantee in each case.

Answer:

> The exact compatible LOOCV shortcut computes the deleted-case score; ERM describes the training optimization; SRM chooses using a declared generalization upper bound. They score different objects and rely on different conditions, so an exact PRESS calculation is not an SRM bound and an ERM minimum is not a final-performance estimate.

We treat the observations as independent pairs (X_i, Y_i) from one population. The data-generating conditional density is $g(y \texttt{ mid } x)$. A regular parametric candidate M_j proposes a conditional density $f_{j(y \texttt{ mid } x, \theta_j)}$ with a finite-dimensional fitted parameter $\theta_j \in \mathbb{R}^{q_j}$. Conditioning on the observed covariates, its log likelihood is

$$\ell_{j(\theta_j)} = \sum_{i=1}^{n} \log f_{j(y_i \texttt{ mid } x_i, \theta_j)}, \quad \hat{\theta}_j = \arg\max_{\theta_j} \ell_{j(\theta_j)}. \tag{20}$$

The integer q_j counts every free quantity fitted by maximum likelihood in that candidate. If a Gaussian model estimates its noise variance, that variance belongs in q_j. The diagonal sum $\mathrm{tr}(H_\lambda)$ from Tutorial 5A is an **effective degrees of freedom** for ridge. It is not automatically the parameter dimension in the standard AIC or BIC derivations. A criterion that substitutes effective degrees of freedom needs its own justified extension.

Keep likelihood score, deviance, and signs separate

We use one convention throughout:

$$D_{\mathrm{fit},j} = -2\ell_{j(\hat{\theta}_j)}, \quad \text{and lower criterion values are better.} \tag{21}$$

This -2 maximized-log-likelihood term is sometimes casually called a deviance. A **saturated model** is a comparison reference flexible enough to reproduce the observed responses exactly; write its maximized log likelihood as ℓ_{sat}. A formal model deviance is often $2\left(\ell_{\mathrm{sat}} - \ell_{j(\hat{\theta}_j)}\right)$. When every candidate uses the same data, likelihood family, and saturated reference, the extra term is constant and rankings agree. Outside that comparison, -2ℓ and deviance should not be silently identified.

IN-CLASS CHECK

Prompt: A Gaussian candidate fits four regression coefficients and an unknown noise variance. Its maximized log likelihood is -121.5. What are D_{fit} and the parameter dimension used by standard AIC? Why is $\mathrm{tr}(H_\lambda)$ not a drop-in replacement?

Hint: Count fitted likelihood quantities, not only regression coefficients.

Answer:

> $D_{\mathrm{fit}} = -2(-121.5) = 243$ and $q = 5$. The ridge trace measures effective response-space flexibility, while standard AIC's q comes from the regular finite-dimensional likelihood expansion. Replacing one by the other changes the criterion and requires a separate derivation.

Training likelihood rewards adaptation. If the candidate list is nested, adding parameters cannot reduce the maximized likelihood. Thus selecting only by D_{fit} drives us toward the richest candidate even when the extra fit is mostly to sample noise. The target must involve a fresh observation.

Before naming a correction, use the fitted likelihoods below. Rank the candidates by training fit, predict which row is most vulnerable to optimism, and write whether a correction should grow or shrink with q_j. Then draw the missing arrow from the training score to a new response at a newly drawn covariate.

M_j	$\ell_{j(\hat{\theta}_j)}$	q_j	fit rank	correction trend	predictive target
M_1	-123	2			training new pair
M_2	-119	5			training new pair
M_3	-118	8			training new pair

Table 6: Pre-commit ledger. Add an arrow in the rightmost gap only after deciding what the training likelihood is trying to predict.

Now formalize the target. For a fitted conditional candidate, define the covariate-averaged conditional Kullback-Leibler discrepancy

$$\mathrm{KL}_{X(g,f_j)} = \mathbb{E}_{X_{\text{new}}}\left[\int g(y \texttt{ mid } X_{\text{new}}) \log \frac{g(y \texttt{ mid } X_{\text{new}})}{f_{j\left(y \texttt{ mid } X_{\text{new}},\hat{\theta}_j\right)}}\,\mathrm{d}y\right]. \tag{22}$$

The contribution involving $g \log g$ is common across candidates. Therefore minimizing expected conditional KL discrepancy is equivalent to minimizing the expected fresh conditional negative log score. With a new pair $(X_{\text{new}}, Y_{\text{new}})$ drawn from the same population,

$$Q_j = \mathbb{E}_{\text{training data}}\mathbb{E}_{(X_{\text{new}},Y_{\text{new}})}\left[-2\log f_{j\left(Y_{\text{new}} \texttt{ mid } X_{\text{new}},\hat{\theta}_j\right)}\right]. \tag{23}$$

Here the inner expectation is over the fresh pair, not only over a response with its covariate silently dropped. For a declared fixed-design study, the analogous target averages the conditional score over those design points.

The training quantity $D_{\text{fit},j}$ is optimistic for Q_j because the same observations chose $\hat{\theta}_j$ and scored it.

Why the AIC correction is $2q$

Let $\theta_j^\star$ denote the parameter value in candidate M_j that minimizes the population conditional negative log score. If the candidate contains the true conditional density, then $f_{j\left(y \texttt{ mid } x,\theta_j^\star\right)} = g(y \texttt{ mid } x)$; otherwise $\theta_j^\star$ is the best population approximation inside that candidate. Expand the fresh score and maximized training log likelihood around this defined reference value. The MLE moves by order $n^{-\frac{1}{2}}$, while quadratic curvature converts that movement into order-one optimism on the -2 log-likelihood scale.

In the ordinary correctly specified regular case, the covariance of the score vector equals the expected negative Hessian of the log likelihood. This is the **information-equality condition**; after standardization its trace is q_j. Training evaluation benefits from the fitted movement and fresh evaluation pays for it, producing total optimism of approximately $2q_j$ on our scale. Hence

$$Q_j \approx \mathbb{E}\left[D_{\text{fit},j} + 2q_j\right] \tag{24}$$

up to candidate-independent constants and smaller-order terms.

This is a large-sample argument. It needs a finite-dimensional regular parametric likelihood, maximum-likelihood fitting, identifiable interior parameters, and consistent parameter counting. Under important misspecification, the simple dimension correction need not be valid; a model-robust replacement requires a separate derivation outside this Tutorial. Standard AIC is not licensed merely because a numerical likelihood exists.

Definition: Akaike information criterion (AIC)

For candidate M_j, the course uses the lower-is-better convention

$$\mathrm{AIC}_j = -2\ell_{j(\hat\theta_j)} + 2q_j. \tag{25}$$

Under the stated regular large-sample conditions, AIC estimates expected out-of-sample conditional KL or negative-log-score discrepancy up to constants common to the candidates.

Claeskens and Hjort write the equivalent higher-is-better score $2\ell_{j(\hat\theta_j)} - 2q_j$. Multiplying the entire criterion by -1 reverses the comparison and leaves the ranking unchanged. Never negate only one column.

Return to the pre-commit ledger. Compute $-2\ell_j$, $2q_j$, and their sum on scratch paper, then commit to the lower-is-better winner before opening the answer.

IN-CLASS CHECK

Prompt: Recompute all three AIC values. Which model wins? If a report instead says "AIC is $2\ell - 2q$ and lower is better," identify the error.

Hint: The book's opposite sign requires the opposite ranking direction.

Answer:

> The values are $250, 248, 252$, so M_2 minimizes AIC. The expression $2\ell - 2q$ is the higher-is-better form. Calling it lower-is-better mixes sign conventions and can reverse the selected model.

The completed decomposition is shown below. A common baseline of 230 has been removed from every bar so that the small criterion differences remain visible.

training likelihood $\rightarrow$ fresh conditional log score for $(X_{\text{new}}, Y_{\text{new}})$

Blue is the shifted training-fit term $-2\ell - 230$; orange is the AIC optimism correction. Assumptions rail: regular finite-dimensional conditional likelihood, maximum-likelihood fitting, a fresh pair from the declared population, and a first-order predictive-risk approximation.

IN-CLASS CHECK

Prompt: Classify each proposed use of standard AIC: (a) three regular Poisson regressions fitted by MLE; (b) an ordinary random forest with "number of leaves" declared to be q; (c) a regular Gaussian regression whose unknown variance is omitted from q; (d) candidate likelihood models with severe nonidentifiability.

Hint: A computable score is not enough; audit the derivation's objects and conditions.

Answer:

> (a) is compatible if the remaining regularity and counting conditions hold. (b) is not standard AIC merely by naming a leaf count. (c) miscounts fitted likelihood parameters. (d) violates a central regularity condition, so the standard expansion cannot be invoked without further theory.

Change the goal, not only the penalty

AIC has supplied a predictive conditional KL target. Suppose instead that the scientific question posits a declared collection of probabilistic models and asks which model has the strongest posterior support after integrating over its parameters. That question introduces a different object.

Before naming the second criterion, compare two proposed rules on the same likelihood table. Rule P is the predictive rule just derived. Rule I strengthens the dimension term as n grows. Compute the three Rule I values, then match P, I, and the familiar SRM card U to the three questions below.

candi-date	$-2\ell_j$	q_j	Rule P: $-2\ell_j + 2q_j$	Rule I: $-2\ell_j + q_j \log n$
M_1	246	2	250	
M_2	238	5	248	
M_3	236	8	252	

question	scientific goal	choose P, I, or U
1	best conditional log score on a fresh pair	
2	identify a represented generating model under a declared regular regime	
3	obtain a stated high-probability risk upper bound over declared classes	

Rule U is the earlier SRM object: empirical risk plus a justified class term. Circle any attempted transfer of U's probability guarantee to P or I.

Table 7: Pre-concept comparison. The arithmetic and scientific question come before the formal name of Rule I.

For candidate M_j, let $\pi_{j(\theta_j)}$ be a parameter prior and let $P(M_j)$ be a model prior. The **marginal likelihood** is

$$p(y_{1:n} \texttt{ mid } x_{1:n}, M_j) = \int \exp\left(\ell_{j(\theta_j)}\right) \pi_{j(\theta_j)} \, \mathrm{d}\theta_j. \tag{26}$$

It averages the likelihood over the prior; it is not the maximized likelihood. Under equal model priors, posterior model ranking follows this integral. With unequal model priors, those prior model probabilities also enter the posterior odds.

Laplace concentration produces the $q \log n$ penalty

For a regular identifiable model with an interior MLE and a prior density that is positive and smooth near it, the log likelihood concentrates in an $n^{-\frac{1}{2}}$-wide neighborhood around $\hat{\theta}_j$. A q_j-dimensional quadratic approximation uses the **per-observation negative Hessian**

$$J_{j(\hat{\theta}_j)} = -\frac{1}{n}\nabla^2_{\theta_j}\ell_{j(\hat{\theta}_j)}. \tag{27}$$

It records the local curvature of the conditional log likelihood. In this regular calculation, assume $J_{j(\hat{\theta}_j)}$ is positive definite, so its determinant is positive. The approximation then gives

$$p\big(y_{1:n} \texttt{ mid } x_{1:n}, M_j\big) \approx \exp\Big(\ell_{j(\hat{\theta}_j)}\Big)(2\pi)^{\frac{q_j}{2}} n^{-\frac{q_j}{2}} \det\Big(J_{j(\hat{\theta}_j)}\Big)^{-\frac{1}{2}} \pi_{j(\hat{\theta}_j)}. \tag{28}$$

Taking -2 logarithms leaves

$$-2\log p\big(y_{1:n} \texttt{ mid } x_{1:n}, M_j\big) = -2\ell_{j(\hat{\theta}_j)} + q_j \log n + O(1). \tag{29}$$

The leading likelihood term is order n, the dimension penalty is order $\log n$, and prior and curvature terms are order one under this regular setup. The prior has not become conceptually irrelevant; BIC has dropped lower-order terms from a prior-dependent marginal-likelihood calculation.

Definition: Bayesian information criterion (BIC)

For candidate M_j and sample size n, the course uses

$$\mathrm{BIC}_j = -2\ell_{j(\hat{\theta}_j)} + q_j \log n, \tag{30}$$

with lower values preferred. It is a large-sample approximation to a marginal-likelihood comparison under the declared regular model and prior regime.

Worked example: One likelihood table, two scientific goals

Reuse the AIC table with $n = 100$, so $\log n \approx 4.605$. The BIC values are

$$M_1 : 246 + 2(4.605) = 255.21, \tag{31}$$

$$M_2 : 238 + 5(4.605) = 261.03, \tag{32}$$

and

$$M_3 : 236 + 8(4.605) = 272.84. \tag{33}$$

AIC chose M_2 for expected predictive conditional log score; BIC chooses M_1 for the stated marginal-likelihood comparison. The disagreement is information about the goal and assumptions, not evidence that one arithmetic rule is universally safer.

IN-CLASS CHECK

Prompt: For $n = 100$, how much maximized log-likelihood improvement must one added parameter supply to overcome the AIC penalty? How much to overcome the BIC penalty?

Hint: Compare changes on the -2ℓ scale.

Answer:

> One extra parameter adds 2 to AIC, so it must improve the maximized log likelihood by more than 1. BIC adds $\log(100) \approx 4.605$, so it must improve the maximized log likelihood by more than about 2.303. These are criterion thresholds, not finite-sample guarantees of correct selection.

Identification is a conditional asymptotic reading

To interpret BIC as identifying a generating model with probability tending to one, state the regime: a declared candidate list, regular identifiable likelihoods, appropriate positive prior mass near the relevant parameters, a large-sample sequence, and a true data-generating model represented in the list. In nested comparisons, the increasing $q \log n$ penalty can remove unnecessary parameters asymptotically. If the true model is absent, the list or parameters are singular, or the prior behavior violates the approximation, that identification statement does not transfer unchanged.

IN-CLASS CHECK

Prompt: A scientist says, "BIC selected M_1, therefore M_1 is the true model." Give the shortest correction that names both the asymptotic and candidate-list conditions.

Hint: Do not replace the claim by "BIC is conservative."

Answer:

> BIC supports consistent identification only under its declared regular large-sample regime, including a represented true model in the candidate list and suitable prior and identifiability conditions. One finite-sample selection does not prove that M_1 is true.

Three questions that must not share a guarantee

Return to Rules P, I, and U. The completed map now names Rule I and reveals the conditions behind each match. Check your assignments before reading the final column.

Method	Scored object	Conditions emphasized	Supported conclusion
AIC	$-2\ell(\hat{\theta}) + 2q$	regular finite-dimensional conditional MLE; large sample	estimated expected fresh conditional KL or log-score discrepancy up to constants
BIC	$-2\ell(\hat{\theta}) + q \log n$	regular marginal-likelihood regime; identification adds true-model-in-list conditions	leading marginal-likelihood ranking; conditional asymptotic identification
SRM	$\hat{R}_{S(h)}$ + class penalty	declared class sequence, loss, confidence level, complexity control	a stated high-probability risk upper bound

Declared goal	Preferred row in example	Reason
predictive conditional log score	M_2 by AIC	smaller corrected predictive criterion
represented-model identification regime	M_1 by BIC	larger asymptotic complexity penalty in marginal-likelihood approximation

Table 8: Target-assumption-guarantee map. The SRM card is deliberately outside the likelihood ranking columns because its probability statement is a different mathematical object.

IN-CLASS CHECK

Prompt: Cross out every unsupported statement: (a) AIC is a distribution-free finite-sample PAC bound; (b) BIC consistency needs no true-model-in-list condition; (c) SRM uses a declared high-probability upper bound; (d) first-order similarity between leave-one-out log score and AIC makes their finite-sample formulas identical.

Hint: A shared anti-overfitting role is weaker than equality of targets or guarantees.

Answer:

Cross out (a), (b), and (d). AIC is likelihood-based and asymptotic; BIC's identification reading is conditional; and first-order asymptotic relation does not make finite-sample formulas identical. Statement (c) is the SRM role established earlier, provided its bound and assumptions are actually declared.

Select inside, evaluate outside

Do this before reading a rule for honest evaluation. For each brief, choose a selection method. In the data-flow cell, draw one arrow through the data allowed to influence the choice and a second arrow to a disjoint reporting box; label the arrows **inner** and **outer**. Finally, name the quantity the last box may honestly report. Leave no pipeline at a slogan such as "use CV."

Brief	Target and family	Data and computation	Choose a method	Draw inner/outer arrows	Reported target
A	predictive squared loss; ridge over 50 fixed λ values	moderate n; fixed-design PRESS identity available			
B	predictive conditional log score; small fixed set of regular Poisson models	MLE reliable; refits possible			
C	identify a represented regular nested likelihood model	large-sample regime; suitable priors and identifiability declared			
D	predictive classification loss; tree pipelines and thresholds	many adaptive choices; no standard regular parameter count			

An information criterion can choose a model without a validation split. It cannot make the training sample fresh again, and it cannot authorize repeated inspection of a final test set. The adaptive procedure includes every model family, hyperparameter, threshold, stopping rule, preprocessing choice, and reporting decision influenced by observed outcomes.

Final performance evidence is honest only when the evaluated observations did not influence any part of that adaptive procedure. Reserve untouched data or new data, or place the whole selection process inside an inner layer and evaluate it with an outer layer.

Trace the arrows before reading the diagnosis. In each outer split, the inner box may run k-fold CV, AIC, BIC, or a declared SRM rule. The outer fold may score only the completed procedure returned by that box.

path	outer training portion		inner selection		outer score
valid	fit and adapt here	$\rightarrow$	choose full procedure	$\rightarrow$	previously untouched outer fold
contaminated	fit candidates	$\rightarrow$	choose after reading outer scores	$\leftarrow$	minimum outer score reported

The green path has no arrow from outer outcomes back into the inner box. The red path reuses evaluation outcomes for choice and then reports the selected minimum as if it were untouched.

Table 9: Inner selection and outer evaluation. Circle every quantity that can change a decision; none may originate in the final scoring layer.

IN-CLASS CHECK

Prompt: A team compares 60 pipelines by inner five-fold CV, chooses one, and reports that minimum inner score. A second team repeats the complete inner procedure inside five outer folds and reports the mean outer loss. What does each number estimate?

Hint: The object being evaluated is either a selected candidate score or the complete adaptive procedure.

Answer:

> The first number is selection evidence: the minimum among reused inner outcomes. It is not an untouched estimate of the full search procedure. The outer-fold mean evaluates the complete inner selection algorithm on observations excluded from every decision in their outer round, subject to the sampling and dependence qualifications of that outer design.

A family resemblance, not one universal equation

All four approaches react to optimism or excessive richness. Their shared role is real, but the mechanism is not a common replaceable penalty.

Rule	What is fit	What is scored	Control mechanism	Typical claim
CV	algorithm on reduced training folds	held-out declared loss	data exclusion and averaging	performance of the fold-trained or selection procedure, with stated transfer
AIC	regular conditional likelihood candidate by MLE	maximized conditional log likelihood	$2q$ predictive optimism correction	expected conditional KL or log-score comparison asymptotically
BIC	regular conditional likelihood candidate by MLE	maximized conditional log likelihood	$q \log n$ marginal-likelihood penalty	conditional marginal-likelihood comparison; conditional identification
SRM	ERM within declared classes	empirical risk plus a justified class term	uniform high-probability complexity control	a stated risk upper bound

IN-CLASS CHECK

Prompt: Why is the slogan "generalization error = training error + complexity penalty" unsafe as a literal identity for all four rows? Give one concrete mismatch involving CV and one involving AIC versus SRM.

Hint: Compare scored objects before comparing algebra.

Answer:

> CV directly scores losses on observations excluded from the corresponding fits; its score is not literally a training error plus the SRM class term. AIC corrects a maximized conditional likelihood toward expected conditional KL or log score under regular asymptotics, whereas SRM uses a declared high-probability bound on risk over classes. The roles resemble one another, but the formulas, assumptions, targets, and guarantees do not match term by term.

Choose from the study, not from a sample-size slogan

A method recommendation should begin with a complete brief:

1. What quantity is the target: predictive loss, predictive conditional log score, or identification in a represented model regime?
2. What candidates are allowed, and is a regular likelihood with consistently counted parameters available?
3. How much data can be reserved, how stable is the algorithm, and what is the cost of refitting?
4. How many choices will adapt to observed scores, and where is the untouched outer evaluation?

The same value of n can lead to different answers because the model family, scientific target, computation, and adaptation depth differ.

IN-CLASS CHECK

Prompt: Compare your four blank pipelines with the method criteria above. For each brief A-D, name the selection rule, describe the inner and outer arrows, and label the final reported target. Then explain why "use BIC when n is large" is not sufficient.

Hint: A complete answer needs four local justifications and one general rejection.

Answer:

A uses the exact compatible ridge LOOCV shortcut for predictive squared loss and still reserves outer evidence. B has a regular likelihood and a predictive conditional log-score target, making AIC natural, while inner CV remains a target-dependent alternative; reported final performance stays outside. C supplies the true-candidate, prior, regularity, and identification regime needed for BIC's intended reading, but predictive claims need their own evaluation. D lacks a standard regular likelihood parameter count and has deep adaptation, so the whole search belongs inside nested CV. Sample size alone says none of this about target, model family, stability, refit cost, or data reuse.

The completed pipelines are the reveal, not the starting point:

Brief	Target and family	Data and computation	Selection rule inside	Evaluation/reporting outside
A	predictive squared loss; ridge over 50 fixed λ values	moderate n; fixed-design PRESS identity available	inner exact LOOCV shortcut, with stability and target qualifications	sealed test or outer layer after choosing and refitting
B	predictive conditional log score; small fixed set of regular Poisson models	MLE reliable; refits possible	AIC or inner CV, chosen according to desired target and diagnostics	untouched conditional log score if a final performance number is reported
C	identify a represented regular nested likelihood model	large-sample regime; suitable priors and identifiability declared	BIC for the conditional identification goal	separate predictive evaluation if predictive performance is claimed
D	predictive classification loss; tree pipelines and thresholds	many adaptive choices; no standard regular parameter count	nested k-fold CV	outer-fold or sealed-test loss of the completed search

Standard language to carry forward

Term or notation	Meaning in this tutorial	Recall cue
maximized log likelihood	The training log likelihood evaluated at its maximum-likelihood estimate.	$\ell_{j(\hat{\theta}_j)}$
parameter dimension	All free likelihood quantities fitted in candidate M_j; not automatically ridge effective degrees of freedom.	q_j
Akaike information criterion	A regular large-sample correction aimed at expected fresh conditional KL or log-score discrepancy.	$-2\ell + 2q$; lower is better
Bayesian information criterion	A regular large-sample marginal-likelihood approximation with a conditional identification reading.	$-2\ell + q \log n$; lower is better
marginal likelihood	Likelihood integrated over the parameter prior, not maximized over parameters.	$\int L(\theta)\pi(\theta)\,\mathrm{d}\theta$

Reconstruct what Chapter 6 may assume

Chapter 6 may rely on exactly three earlier ideas, in this order. They are pass-through knowledge, not new content introduced by this Tutorial.

No.	Available idea	Reconstruction
1	realizable PAC learning	State the high-probability small-risk goal under a represented zero-risk hypothesis and sample-size control.
2	convex function background	Recognize the chord inequality and the local-to-global optimization consequence already established in Chapter 1.
3	empirical risk minimization	Identify the hypothesis minimizing the declared empirical loss over its class.

IN-CLASS CHECK

Prompt: Without looking back, reconstruct the three-item Chapter 6 entry in order and say what this Tutorial does **not** authorize Chapter 6 to assume about AIC, BIC, or final evaluation.

Hint: Separate entry knowledge from this chapter's assessed model-selection knowledge.

Answer:

> The order is realizable PAC learning, convex-function background, and ERM. This Tutorial does not make AIC or BIC distribution-free PAC bounds, does not make their targets interchangeable with ERM, and does not permit any final-evaluation data to influence the procedure it evaluates. Those distinctions remain Chapter 5 knowledge, while the next chapter receives only the three reconstructed earlier ideas.

Source note

The historical Week 5 Learning Sheet fixes the official scope of information criteria and model-selection synthesis, printed/PDF pp. 25-41. Its sign- inconsistent example table, automatic sample-size cutoffs, ordinary-k-NN PRESS claim, universal method recommendations, literal common-penalty equation, and unqualified statements that every method estimates the same test error are not reused.

The AIC definition, covariate-averaged conditional KL target, likelihood-optimism expansion, opposite-sign book convention, and first-order relation to leave-one-out log score follow Claeskens and Hjort, Sections 2.1-2.4 and 2.9, printed pp. 22-42 and 51-55 (PDF pp. 40-60 and 69-73). The standard $2q$ form is kept inside its regular parametric scope; misspecification is flagged as requiring a separate model-robust derivation rather than silently replacing that derivation by parameter dimension.

The BIC formula and Laplace marginal-likelihood construction follow Claeskens and Hjort, Chapter 3, especially equation (3.1) and Section 3.2, printed pp. 69-96 (PDF pp. 87-114); the normalized negative Hessian used in the determinant is defined on printed p. 80 / PDF p. 98. The prediction-versus-identification distinction and its conditional asymptotic regime are checked against Chapter 4, printed pp. 99-114 (PDF pp. 117-132).

The SRM and cross-validation objects and their distinct guarantees follow Mohri, Rostamizadeh, and Talwalkar, Sections 4.3-4.6, printed pp. 64-73 (PDF pp. 81-90). The final train-selection-test boundary follows Shalev-Shwartz and Ben-David, Section 11.2.5, printed/PDF p. 150. These sources support a comparison of roles; they do not support turning AIC, BIC, CV, and SRM into one universal algebraic identity or transferring a PAC guarantee to an information criterion.